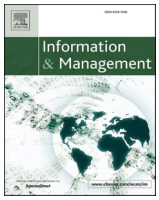
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Leveraging structural IT capabilities to promote novelty and efficiency in business model design: A knowledge-based view

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ABSTRACT

Enterprises are confronted with the challenge of designing business models grounded in information technology (IT). In addressing this concern, prior literature has highlighted the significance of intra-organizational IT capabilities in the business model design (BMD), yet insufficient attention has been devoted to inter-organizational IT capabilities. Employing a knowledge-based view to address this research gap, this study explores how enterprises leverage structural IT capabilities to foster novelty-centered business model design (NBMD) and efficiency-centered business model design (EBMD). Using matched survey data from 180 Chinese manufacturing firms, our findings indicate a positive correlation between IT reconfiguration and NBMD, and a positive correlation between IT integration and EBMD. Additionally, polynomial regression and response surface analysis show that maintaining a high proportion of IT reconfiguration or integration is required to optimize NBMD. For facilitating EBMD, a high proportion of IT integration must be maintained, and a high proportion of IT reconfiguration has a comparatively lesser impact. Our research results offer theoretical and practical implications for how to develop structural IT capabilities to support BMD.

1. Introduction

In the digital era, enterprises confront the challenge of leveraging information technology (IT) to redesign their business models for new opportunities in a dynamic environment [71]. Business model design (BMD) serves as a strategic approach enabling enterprises to adapt to environmental changes [44,61,62,79], which outlines the process by which an enterprise engages with stakeholders to create and capture value [1,71,79]. To sustain these transactions across organizational boundaries in the digital era, enterprises must enhance their IT structure, employing IT reconfiguration and IT integration to ensure knowledge sharing and value co-creation [32,70–72,81,97]. For example, IBM Watson Health captures healthcare data better due to reconfiguring its information systems and establishing connections with healthcare providers, research institutions, and patient databases. This data is then applied to develop advanced AI diagnostic tools and treatment options, enabling Watson Health to transform its business model from a traditional hardware and software company to a leader in healthcare analytics and personalized medicine. While the benefits of IT are undeniable, the limited resources of most enterprises present challenges in achieving the bimodal IT reconfiguration and integration necessary for driving transformations [31,77]. Consequently, it is imperative for enterprises to decide prudently how to allocate resources for IT reconfiguration and integration to propel BMD.

In the business model literature, categorized by distinct design domains of transaction methods and value creation, business model design (BMD) can be segmented into novelty-centered BMD (NBMD) and efficiency-centered BMD (EBMD; [98–100]). Specifically, NBMD concentrates on developing novel and differentiated approaches to stakeholder value creation. This requires enterprises to gain insights into market trends, consumer preferences, and technological innovations, continuously acquiring and updating knowledge about market dynamics, emerging technologies, and innovative approaches [15,36]. Unlike NBMD, EBMD creates value in an efficient, low-cost, and low-risk manner [98–100]. It emphasizes process optimization and cost control, necessitating knowledge of supply chain management, production efficiency, and logistics [15,36]. Previous studies have demonstrated that IT capabilities serve as knowledge enablers in promoting BMD [15,27,36,71]. However, the majority of published studies have concentrated on intra-organizational IT capabilities, examining the impact of IT management capabilities [15] or IT human capital [36] on BMD, while largely overlooking the role of inter-organizational IT capabilities.

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Within the inter-organizational systems literature, structural IT capabilities, encompassing IT reconfiguration and IT integration, are recognized as crucial technological capabilities for overseeing transactional activities with stakeholders [30,70]. In particular, IT reconfiguration involves constructing modular information system components with standardized interfaces and common protocols, while IT integration emphasizes the seamless connection of systems between enterprises [65,70,95]. Enterprises must develop IT reconfiguration and integration to ensure long-term prosperity. IT reconfiguration can establish connections with new partners, expanding access to knowledge and exploring new market opportunities, while IT integration can facilitate the integration and sharing of knowledge with existing partners [70,95]. A recent empirical study on organizational agility has shown that maintaining a balance between IT reconfiguration and integration is an important enabler [65]. Other research has shown that allocating resources evenly to the development of IT capabilities is often suboptimal; for most enterprises, concentrating more on one is a preferable strategy [58]. Consequently, enterprises must ascertain the optimal proportion, namely the proportional balance, of IT reconfiguration and integration. However, the literature lacks a comprehensive discussion of how a proportional balance of efforts toward structural IT capabilities impacts BMD, and empirical support is also deficient.

To fill the above research gap, we investigate the impact of IT reconfiguration and integration, along with their proportional balance effect, on BMD, based on the knowledge-based view (KBV). The KBV formulates theories about the conditions under which enterprises effectively assimilate external knowledge and the resulting impact on enterprises' outcomes [16,67,81]. Additionally, the KBV emphasizes that while enterprises need to search for external knowledge, they must also integrate it into their operations and innovation processes [51]. Building upon this theory, we present a theoretical model elucidating how structural IT capabilities impact BMD. We take a variety of approaches to validate our theoretical model, using matched survey data from 540 senior executives (CEOs, R&D managers, and marketing managers) from 180 manufacturing SMEs in China.

The empirical findings make three distinct contributions to the literature. First, we enhance the comprehension of inter-organizational IT capabilities as pivotal antecedents in BMD by offering empirical evidence on how structural IT capabilities serve as knowledge initiators to foster BMD. Second, we deepen the understanding of the relationship between IT capabilities and BMD by introducing a proportional balance perspective. Our results suggest that enterprises can enhance the novelty of BMD by having a high proportion of IT reconfiguration or integration, as well as a high proportion of IT integration to promote the efficiency of BMD. Third, this study introduces a proportional balance perspective into the research question of IT promoting knowledge management, enriching the KBV literature stream. Previous studies have emphasized the direct effect of IT on knowledge management, but how the proportional balance of IT capabilities affects knowledge acquisition and assimilation remains unclear. Our study contributes to the KBV literature by emphasizing this novel perspective and providing empirical evidence.

2. Theoretical basis and hypotheses

2.1. KBV of the firm

The KBV defines an enterprise as an institution that actively seeks and integrates knowledge. This theory emphasizes the fundamental effects of knowledge on innovation and performance, particularly the transfer of knowledge within and between enterprises [29,55,67,82]. The KBV holds significant implications for our research, since a business model imposes stringent requirements for knowledge [76]. Strategic entrepreneurship scholars highlight that business model design—aiming to attain a competitive advantage—often requires novel knowledge combinations [67,82]. Moreover, business models are increasingly reliant on external knowledge, especially from suppliers, customers, universities, and research institutions [19,39]. To facilitate the exchange of knowledge needed for business models, the evolution of digital technology provides data access systems and network connectivity technologies that break down the boundaries of enterprises by eliminating communication barriers [81]. In the literature, IT is regarded as a powerful tool for locating and assimilating external knowledge [5,9,21,37,40,45,47,68,81]. Thus, grounded in the KBV, we expect that inter-organizational IT capabilities influence BMD by fostering the acquisition and integration of external knowledge.

In addition, a fundamental tenet of the KBV is that, as enterprises seek external knowledge, they must also integrate it into their operations and innovation processes [51]. Based on this premise, the significance of fostering inter-organizational IT capabilities for locating and integrating external knowledge has become evident. However, in practical business scenarios, enterprises frequently encounter challenges in allocating their IT investments between building external knowledge search and integration capability. As an illustration, J.C. Penney, an American department store chain, revamped its traditional system and integrated systems throughout the supply chain to facilitate the shift to e-commerce. However, they did not consider resource allocation effectively, ultimately resulting in failure. In such situations, the majority of enterprises confront the challenge of exploring suitable input allocation strategies that account for both locating and integrating external knowledge.

2.2. BMD and structural IT capabilities

A business model delineates how a focal firm and its stakeholders engage in transactions and generate value [1,79,98–100]. This aspect is of paramount importance for sustaining competitive advantage. Grounded in distinct design fields encompassing transaction methods and value creation, BMD can be categorized into NBMD and EBMD [98–100]. Specifically, NBMD generates value through innovative means that may necessitate a heightened level of novelty in the knowledge portfolio held by the enterprise. NBMD centers on constructing novel and distinctive approaches to create value for stakeholders. Enterprises must continually garner insights into market trends, consumer preferences, and technological innovations, ensuring the acquisition and updating of knowledge about market dynamics, emerging technologies, and innovative approaches [15,36]. Inconsistently with the NBMD value creation goal, EBMD generates value by emphasizing efficient, low-cost, and low-risk operations [98–100]. It concentrates on process optimization and cost control, demanding profound knowledge of supply chain management, production efficiency, and logistics [36].

Prior research has highlighted the necessity of novel knowledge combinations for business models [67,82]. From a knowledge management perspective, BMD can be regarded as a process of consolidating and generating knowledge [14,92]. Extensive research has additionally highlighted that high-yield knowledge exists not solely within enterprises but also externally [2,28,53,88]. There are other scholars who argue that most enterprises need to seek external knowledge sources to design their business models, integrating and adapting them to internal capabilities [4,20]. Their business models can become inseparable from their networks of partners [33,60,62,83]. Recent literature advocates additional research on this issue [27,36,93].

IT works as a powerful tool for knowledge diffusion [45,64,74,94]. There are a number of studies exploring the role of IT in facilitating business models [15,27,36,71]. For instance, Sorescu [75] demonstrated that enterprises can leverage information networks and big data to acquire internal and external knowledge, enabling the creation of new business models. Cozzolino et al. [17] also emphasized that digital transformation can help enterprises update business models to seize external opportunities more effectively. However, most of the above studies have taken the perspective of intra-organizational IT

capabilities, and the discussion of inter-organizational IT capabilities and business models is still insufficient, especially of how the proportional balance of IT capabilities will affect enterprise knowledge management is still unclear.

2.3. Separate effects of IT reconfiguration and IT integration on BMD

Structural IT capabilities, encompassing IT reconfiguration and IT integration, are recognized as crucial technological capabilities for effectively managing enterprise knowledge, which align with comprehensive knowledge location and internal integration [32,70–72,95]. First, IT reconfiguration involves constructing modular information system components with standardized interfaces, common protocols, and other features to ensure the compatibility of information systems [70]. IT reconfiguration broadens an enterprise's access to knowledge primarily by establishing connections with new partners, which facilitates the exploration of new market opportunities and technology trends [65,95]. Second, IT integration emphasizes the seamless connection of systems between enterprises [8,32]. This helps enterprises utilize the knowledge and resources of partners and adjust the transaction process in a timely manner to realize joint value creation. In contrast to IT reconfiguration, IT integration concentrates more on integration and sharing of knowledge among existing partners. This enables enterprises to manage their supply chains effectively, reduce information asymmetry, and enhance the accuracy of decision-making [70].

We propose that both IT reconfiguration and IT integration can propel NBMD. First, IT reconfiguration can enhance an enterprise's system to accommodate a greater number of new partners [3,66,97], including new suppliers and customers. It enables access to complementary resources and knowledge [23], facilitating swift sharing and access to information on market and technological innovations through compatible systems with external partners. By acquiring new knowledge from a broader spectrum of suppliers or customers, enterprises extend the coverage and richness of their enterprise knowledge base [70,74]. This enables them to innovate business processes more effectively [80,82]. The new knowledge obtained from new partners can be applied to enhance the novelty of BMD.

Additionally, we propose that IT integration can enable NBMD. IT integration facilitates a deeper understanding of market trends and consumer needs and assists enterprises in enhancing their knowledge base to explore new business opportunities, develop novel transaction structures, and devise innovative value-creation methods. This differentiation extends to both their existing and emerging business models [15]. A seamlessly integrated IT system facilitates the exchange of business information and embedding of business activities and processes between enterprises and their partners [35,73]. This, in turn, assists enterprises in internally transforming external knowledge. Through exchanging knowledge, enterprises can generate new sources of value, consequently enhancing NBMD [6,34]. We hypothesize the following:

Hypothesis 1a: IT reconfiguration relates positively to the degree of NBMD.

Hypothesis 1b: IT integration relates positively to the degree of NBMD.

IT reconfiguration can also promote EBMD by facilitating the dynamic alignment of business operations with partners [70]. IT reconfiguration significantly reduces the cost of information system redevelopment, thereby contributing to the efficient management of internal processes and the accurate analysis of data. Additionally, IT reconfiguration can ensure that the interfaces or protocols of an enterprise's information systems are more likely to be accepted by potential new partners, including potential suppliers of existing production lines [71]. With the addition of new partners, the steps for acquiring new information and knowledge are simplified, effectively streamlining the transaction process [43]. Consequently, enterprises improve efficiency and reduce cost, aligning with the requirements of enterprise EBMD.

Accordingly, IT integration resolves disparities in data exchange standards or interfaces and facilitates the integration of partner knowledge [11]. IT integration assists enterprises in promptly tracking the needs of cooperating enterprises, effectively mitigating the risks stemming from transaction uncertainty. This, in turn, optimizes inventory management and improves cost efficiency [8,22,64]. As an illustration, by establishing a highly integrated supply chain management system, Walmart has achieved swift knowledge exchange with its partners [10,46]. This has significantly diminished the bullwhip effect resulting from inadequate data exchange [54,90]. We hypothesize the following:

Hypothesis 2a: IT reconfiguration relates positively to the degree of EBMD.

Hypothesis 2b: IT integration relates positively to the degree of EBMD.

2.4. Proportional balance of IT reconfiguration and integration

Given the variations between NBMD and EBMD in the demand for different types of knowledge, enterprises require a combination of structural IT capabilities to augment the business value of IT. Balanced structural IT implies that the resources invested in IT reconfiguration and integration receive equal effort. With the enhancement of balanced total effort between IT reconfiguration and integration, enterprises can swiftly access the information systems of potential partners to acquire new knowledge and resources [3] and update their knowledge bases [23]. Furthermore, enterprises can rapidly integrate knowledge into their own business processes, diminishing reliance on interactive memory for supporting knowledge integration [81] and ensuring transaction efficiency [8,22,64]. In contrast to a balanced high effort, a balanced low effort faces challenges in adapting to changes such as interface and protocol upgrades driven by technological progress. This limitation confines enterprises to existing partner groups and poses a risk of partner group reduction [70].

However, we are also aware that recent studies have pointed out that an unbalanced IT resource allocation strategy may be better for some enterprises. If IT resources meet firms' current strategic business needs, IT will create value [13,15,36,57,58]. Especially when facing limited resources, enterprises must exercise caution in deciding on investment allocation. To better reflect this reality, research by Rai and Tang [70] and Liang et al. [58] provides insights, leading us to propose a proportional (rather than even) balance approach to address the shortcomings of previous research that cannot distinguish between high and low proportions of IT reconfiguration and integration (e.g., [65]).

NBMD generates value for enterprises in innovative ways, manifesting as new trading methods, novel value-creation participants, or innovative connectivity methods [98]. IT reconfiguration offers an efficient means for enterprises to obtain new knowledge from potential partners [66,70,97], allowing enterprises to establish IT connections with partners with whom they have not traded before [71]. Additionally, IT reconfiguration establishes bridges to a broad spectrum of potential partners and fosters the generation of new ideas in product services, manufacturing operations, and distribution [78]. Consequently, IT integration concentrates on profound knowledge integration with its existing partners [48,49], enhancing the enterprise knowledge base and establishing mechanisms for collaborative innovation. While both IT reconfiguration and IT integration may support NBMD, IT reconfiguration, in comparison to IT integration, can introduce more novel elements to the business model, so we hypothesize the following:

Hypothesis 3a: The degree of NBMD is higher when the total balanced effort between IT reconfiguration and integration is higher.

Hypothesis 3b: The degree of NBMD is higher when the proportion of IT reconfiguration effort is higher than that of IT integration.

In contrast to NBMD, EBMD generates value for enterprises by streamlining transaction processes, thereby diminishing costs and risks [98,99]. A highly integrated information system diminishes the information asymmetry and coding dislocation between the enterprise and its

suppliers or customers, enhancing the informed decision-making of both parties and achieving a tight junction of knowledge resources [71]. Consequently, IT integration assists enterprises in timely tracking of the needs of cooperative enterprises, effectively mitigating the risks caused by transaction uncertainty, and significantly enhancing transaction efficiency [8,22,64]. Thus, IT reconfiguration delivers flexible information technology services [70], significantly enhancing the compatibility of an enterprise's information systems and ensuring flexibility in the provisioning and allocation of resources [26]. While IT reconfiguration can help enterprises align IT swiftly with business processes [57], it can also entail unforeseen risks and high maintenance costs [38]. Therefore, we hypothesize the following:

Hypothesis 4a: The degree of EBMD is higher when the total balanced effort between IT reconfiguration and integration is higher.

Hypothesis 4b: The degree of EBMD is higher when the proportion of IT integration effort is higher than that of IT reconfiguration.

3. Empirical study

3.1. Sample and data

We conducted a questionnaire survey in the Yangtze River delta of China to validate our research model empirically. China serves as a crucial global manufacturing base, and the Yangtze River delta region boasts the nation's highest per capita GDP and a well-established manufacturing industry [61,63,96]. The application of IT has established a solid foundation for manufacturing enterprises to establish connections with other enterprises [63], facilitating the active integration of partners into a supply chain [42]. Consequently, the Yangtze River delta offers an ideal environment for undertaking inter-organizational IT research.

In 2015, the Chinese government unveiled the "Made in China 2025" (MIC2025) plan [18] to encourage manufacturing small and medium-sized enterprises (SMEs) to leverage IT to enhance their business models [56,85]. Consequently, we gathered matching data between 2016 and 2017. We chose manufacturing enterprises, specifically SMEs, as our research sample for two primary reasons. First, IT reconfiguration and integration are both very important to the business model of manufacturing enterprises. However, when compared with established enterprises, SMEs frequently encounter financial and technological constraints, so that harnessing their IT capabilities for promoting BMD poses a more significant challenge for them [86]. Consequently, selecting SMEs holds significant implications for discerning the crucial IT capabilities for BMDs to optimize the allocation of limited resources. Second, we selected SMEs as our research sample because they constitute over 90 % of market entities in China, thus providing substantial representativeness.

Before distributing the questionnaire, we cooperated with the local government to obtain a list of approximately 1200 small and medium enterprises. Using the contact information of these enterprises, we invited their managers to participate in our online survey voluntarily. The study focuses on manufacturing SMEs to ensure that business models are comparable across enterprises. In addition, to make our sample representative, we used random sampling and selected 600 enterprises. Each enterprise's CEO answered the questions on BMD, given that the CEO has a clear and comprehensive view of BMD. The questions on structural IT capability were answered by R&D managers, and market managers answered the questions on external environmental uncertainty, including technological turbulence, market turbulence, and competitor turbulence. In the end, we collected 180 questionnaires for data analysis by matching questionnaires to CEOs, R&D managers, and market managers, representing a response rate of 30 %. Table 1 shows the sample data collected from enterprises.

Table 1
Sample statistics (N= 180).

	N	Percentage (%)
Firm age		
≤8	80	44.44
8–15	72	40.00
>15	28	15.56
Firm size[a] (thousand)		
≤1000	58	32.22
1000–5000	74	41.11
5000–10,000	29	16.11
>10,000	19	10.56
Industry		
Consumer products	39	21.67
Petroleum and chemical	42	23.33
Machinery	9	5.00
Electronics	48	26.67
Mineral	42	23.33

[a] Total firm assets.

3.2. Measures

We identified published scales of the relevant variables and adapted them to the context of our study. Each indicator was measured using a five-point Likert scale. To ensure conceptual equivalence, we facilitated the translation of the English questionnaire into Chinese and sought feedback from two native English scholars who provided reverse translations. We also invited three scholars with extensive research in the field of information systems and strategic management to evaluate our questionnaire, and we revised the questionnaire according to their suggestions. In Table 2, we provide an overview of the measures for relevant variables, accompanied by their respective factor loadings.

Structural IT capabilities. Based on the concept of structural IT capabilities constructed by Rai and Tang [70], we designed eight items to measure structural IT capabilities. The four items for measuring IT reconfiguration reflect the compatibility of information systems well to ensure that enterprise search can search for and expand external knowledge sources, such as simply expanding to access new systems or being accommodated by other systems. For IT integration, the four items effectively reflect the degree of close connection between the enterprise information system and partners, such as quickly obtaining information from the supplier's system.

Business model design. The scale for BMD was developed by Zott and Amit [98], and later scholars tested this scale and developed simplified versions based on it. The reliability and validity of the simplified version have also been proven. We used Brettel's [12] version of the scale, which has been used in published articles [44,60,62]. This measurement of NBMD reflects novelty well. For example, we focus on describing new forms of transactions between stakeholders, such as new methods of integrating products and services. To measure EBMD, we focus on describing transaction costs and the efficiency of business processes. For example, the business model of an enterprise can facilitate rapid transactions and effectively reduce costs.

Control variables. To control for the potential interference of various factors with enterprise BMD, we added variables such as the age and size of the enterprise into our hypothesis-testing model. First, compared with new start-ups, long-established enterprises tend to have more management experience, potentially influencing BMD tendencies. Second, we use the assets of the enterprise to measure its size. The assets of an enterprise may indicate the extent of the enterprise's dominance in the industry, which is likely to affect inter-enterprise relationship management. Next, we introduce five dummy variables to measure the industry: consumer goods, petroleum and chemical products, machinery, electronic products, and minerals industries. Finally, we also control for the environmental uncertainty in which the enterprise operates. External environmental uncertainty includes technological turbulence, market turbulence, and competitor turbulence, and we measure it using a scale

Table 2
Measurement items and confirmatory factor analysis results.

Construct and items	Loading
NBMD (Cronbach's α = 0.87; CR = 0.88; AVE = 0.60)	
1. Our business model offers new links between stakeholders (such as customers, suppliers, etc.).	0.682
2. The richness (i.e., quality and depth) of some of the links between participants is novel.	0.730
3. In our industry, we are a pioneer with our business model.	0.696
4. There are other important aspects of the business model that make it novel.	0.869
5. Overall, our business model is novel.	0.882
EBMD (Cronbach's α = 0.88; CR = 0.88; AVE = 0.60)	
1. Transactions are simple from the user's point of view.	0.715
2. Our business model enables a low number of errors in the execution of transactions.	0.737
3. Our business model enables participants to make informed decisions.	0.642
4. Our business model enables fast transactions.	0.851
5. Our business model, overall, offers high transaction efficiency.	0.886
IT reconfiguration (Cronbach's α = 0.95; CR = 0.95; AVE = 0.83)	
1. Our information system is easily adapted to include new suppliers/customers.	0.915
2. Our information system can be easily extended to accommodate new applications or functions.	0.933
3. Our information system employs standards that are accepted by most current and potential suppliers/customers for this product line.	0.909
4. Our information system consists of modular software components, most of which can be reused in other business applications.	0.888
IT integration (Cronbach's α = 0.95; CR = 0.95; AVE = 0.82)	
1. Our information system easily accesses data from our suppliers' systems/customers' systems.	0.907
2. Our information system provides seamless connection between our suppliers' systems/customers' systems and our systems (e.g., forecasting, production, manufacturing, shipment, etc.).	0.932
3. Our information system has the capability to exchange real-time information with our suppliers/customers.	0.800
4. Our information system easily aggregates relevant information from our suppliers' databases/customers' databases. (e.g., operating information, business customer performance, and cost information).	0.975
Technological turbulence (Cronbach's α = 0.84; CR = 0.85; AVE = 0.59)	
1. The technology in our industry is changing rapidly.	0.785
2. It is very difficult to forecast where the technology in our industry will be in the next two to three years.	0.697
3. A large number of new product ideas have been made possible through technological breakthroughs in our industry.	0.798
4. The technological changes in this industry are frequent.	0.785
Market turbulence (Cronbach's α = 0.70; CR = 0.75; AVE = 0.51)	
1. In our kind of business, customers' product preferences change quite a bit over time.	0.811
2. We are witnessing demand for our products and services from customers who have never bought them before.	0.674
3. It is very difficult to predict any changes in this marketplace.	0.638
Competitor turbulence (Cronbach's α = 0.87; CR = 0.87; AVE = 0.63)	
1. Competition in our industry is cutthroat.	0.797
2. There are many "promotion wars" in our industry.	0.817
3. Price competition is a hallmark of our industry.	0.717
4. One hears of a new competitive move almost every day.	0.828

developed by Wilden and Gudergan [89].

4. Analysis and results

4.1. Reliability and validity

To test the validity of the constructs, we employed various testing methods. First, the minimum values of Cronbach's α and CR of all variables were 0.70 and 0.75, exceeding the standard of 0.70 [52]. This suggests that our questionnaire demonstrates good reliability (see Table 2). Second, the CFA test was carried out using AMOS 22.0, and the test results show that the model fits the data well ($\chi^2/df = 601.586/356 = 1.690$, CFI = 0.936, IFI = 0.937, NFI = 0.858, RMSEA = 0.062, SRMR = 0.054), signifying strong convergent validity. Finally, we also found that all AVE scores exceeded the recommended threshold of 0.50 (see Table 2), while their square roots surpassed the correlation coefficients with other constructs (see Table 3).

4.2. Common method bias

To minimize the impact of common method bias, measures were taken before data were collected for analysis. Specifically, we placed the relevant items of independent and dependent variables on separate pages to minimize the consistency of interviewees' thinking [69]. Moreover, we collected measurements of different variables from CEOs, R&D managers, and marketing managers to reduce this bias. Finally, we used statistical analysis to assess common method bias. We used Harman's single factor test [59], and the results showed that five constructs have eigenvalues larger than 1, accounting for 72.83 % of the total variance. The first of these constructs explains 27.75 % of the variance, satisfying the recommendation that it be below 40 %. In conclusion, common method bias is unlikely to threaten the results.

4.3. Hypothesis testing

To elucidate this model, we utilized two distinct methodologies. First, following the recommendations of previous scholars, we applied the hierarchical regression method to test our main effect. This method calculates the coefficients of the corresponding main effect variables. Second, we performed polynomial regression and response surface analysis to examine the impact of the proportional balance effect between IT reconfiguration and integration on BMD.

4.3.1. Main effect of IT reconfiguration and integration

First, we utilized SPSS 24.0 to run a hierarchical regression model. Following the guidance of previous scholars [7], we used a hierarchical regression method to test our hypotheses. To mitigate the potential impact of multicollinearity on our research, we normalized each variable. The statistical results show that the maximum value of the variance inflation factor is 2.439, falling below the suggested upper threshold of 5. This indicates that multicollinearity is not a major concern in our study.

In our hypotheses, H1a and H1b predict that IT reconfiguration and integration impact NBMD positively. The results of Model 2, shown in Table 4, show that IT reconfiguration is related positively to NBMD ($\beta = 0.291$, $p < 0.01$), but IT integration is not significant ($\beta = 0.152$, $p > 0.05$). Therefore, only H1a is supported. H2a and H2b posit that IT reconfiguration and integration promote EBMD. In Table 4, the Model 4 results show that only IT integration has a significant positive effect on EBMD ($\beta = 0.269$, $p < 0.01$), so H2b is supported while H2a is not.

4.3.2. Polynomial regression and response surface analysis

We use polynomial regression and response surface analysis methods to provide empirical evidence regarding a proportional balance between IT reconfiguration and integration. According to Edwards [25] and previous information systems research [84], the functional form of polynomial regression is

$$z = \beta_0 + \beta_1 x + \beta_2 y + \beta_3 x^2 + \beta_4 xy + \beta_5 y^2 + e$$

where z represents the dependent variable (including NBMD and EBMD), x represents IT reconfiguration, and y represents IT integration. We mean-center IT reconfiguration and IT integration to reduce multicollinearity [24]. To visualize the proportional balance effect for a more intuitive analysis, we use regression coefficients to construct a response surface, which reveals a mesh image with the perfect balance line $y = x$ and the perfect imbalance line $y = -x$. The surface slope along the

Table 3
Sample statistics and correlations.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. NBMD	**0.775**													
2. EBMD	0.619***	**0.775**												
3. IT reconfiguration (ITR)	0.389***	0.303***	**0.911**											
4. IT integration (ITI)	0.345***	0.363***	0.607***	**0.906**										
5. Technological turbulence (TT)	0.100	0.008	0.157*	0.063	**0.768**									
6. Market turbulence (MT)	−0.011	−0.126	0.021	−0.147*	0.668***	**0.714**								
7. Competitor turbulence (CT)	0.043	−0.069	0.022	−0.039	0.659***	0.674***	**0.794**							
8. Firm age	−0.067	−0.087	−0.217**	−0.238**	0.053	0.167*	0.148*							
9. Firm size	−0.090	0.002	−0.067	−0.0171*	0.028	0.040	0.052	0.066						
10. Consumer products [a]	−0.060	0.026	−0.084	−0.076	0.065	0.060	−0.050	−0.137	−0.094					
11. Petroleum and chemical [a]	−0.004	0.035	0.043	−0.040	−0.092	−0.131	−0.024	0.130	−0.033	−0.290***				
12. Machinery [a]	−0.007	−0.123	0.038	−0.052	−0.068	−0.002	−0.062	−0.181*	0.094	−0.121	−0.127			
13. Electronics [a]	0.051	0.002	0.039	0.127	0.077	0.048	0.151*	0.013	−0.107	−0.317***	−0.333***	−0.138		
14. Mineral [a]	0.013	0.001	−0.022	0.008	−0.016	0.023	−0.053	0.083	0.188*	−0.290***	−0.304***	−0.127	−0.333***	
M	3.955	4.156	3.689	3.697	3.735	3.672	3.821	10.18	7.717	0.220	0.230	0.050	0.270	0.230
SD	0.564	0.437	0.704	0.749	0.696	0.629	0.677	5.311	1.264	0.413	0.424	0.219	0.443	0.424

Notes: The diagonal elements (boldface values) are the square roots of the AVEs. Unadjusted correlations are below the diagonal.
[a] Dummy variable.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

Table 4
Hierarchical regression results.

	NBMD		EBMD	
	Model 1	Model 2	Model 3	Model 4
Firm age	−0.063	0.037	−0.086	0.008
Firm size	−0.098	−0.056	0.027	0.078
Consumer products	−0.085	−0.021	0.006	0.071
Petroleum and chemical	−0.021	−0.019	0.021	0.040
Machinery	−0.007	0.006	−0.130	−0.105
Electronics	0.012	0.020	−0.001	0.008
Technological turbulence (TT)	0.183	0.062	0.156	0.044
Market turbulence (MT)	−0.130	−0.076	−0.187	−0.105
Competitor turbulence (CT)	0.020	0.050	−0.041	−0.026
IT reconfiguration (ITR)		0.291**		0.151
IT integration (ITI)		0.152		0.269**
R^2	0.039	0.179	0.052	0.175
R^2 change		0.140		0.122
F change		14.312***		12.464***

** $p < 0.01$.
*** $p < 0.001$.

perfect balance line refers to the impact on BMD when the degrees of IT reconfiguration and integration effort are the same. The surface curvature along this line indicates whether there is a curved relationship between the balance of IT reconfiguration and integration and BMD. The perfect imbalance line indicates that there are variations in the degrees of IT reconfiguration and integration, but their summed effort is certain. Illustrating with an example, enterprises may have different investment ratios for the two types of structural IT capabilities, such as ratios of 2:8 or 7:3. The surface slope along the perfect imbalance line refers to the effect of the variance direction on the BMD. The surface curvature along this line refers to how the degree of variance between IT reconfiguration and integration will affect BMD. Table 5 presents the results of the polynomial regression, and we use the polynomial regression coefficients to construct the response surface images of NBMD and EBMD, depicted in Figs. 1 and 2.

H3a and H3b posit that a proportionally balanced effort for IT reconfiguration and integration drives NBMD. Observations of the response surface results show that the slope of the response surface along the perfect balance line (β = 0.421, 90 % CI [0.294, 0.544]) is positive and significant. This result suggests that NBMD is higher (i.e., achieves better success) when the balanced effort is high than when the balanced effort is low, validating H3a. In addition, the significant curvature (β = 0.163, 90 % CI [0.022, 0.252]) suggests that the relationship between balance effort and NBMD cannot be characterized simply as a

Table 5
Polynomial regression results.

Variables	NBMD	EBMD
Constant	3.753***	3.999***
Firm age	0.006	0.002
Firm size	−0.031	0.026
Consumer products	−0.034	0.057
Petroleum and chemical	−0.025	0.045
Machinery	0.041	−0.135
Electronics	0.065	0.054
Technological turbulence (TT)	0.095	0.057
Market turbulence (MT)	−0.105	−0.106
Competitor turbulence (CT)	0.052	0.001
IT reconfiguration (ITR)	0.227**	0.047
IT integration (ITI)	0.194*	0.222***
ITR^2	0.248*	−0.047
ITR × ITI	−0.273*	0.105
ITI^2	0.188	0.124*
R^2	0.262	0.272
Adj R^2	0.199	0.210
R^2 change for the three quadratic terms	0.083***	0.097***

* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

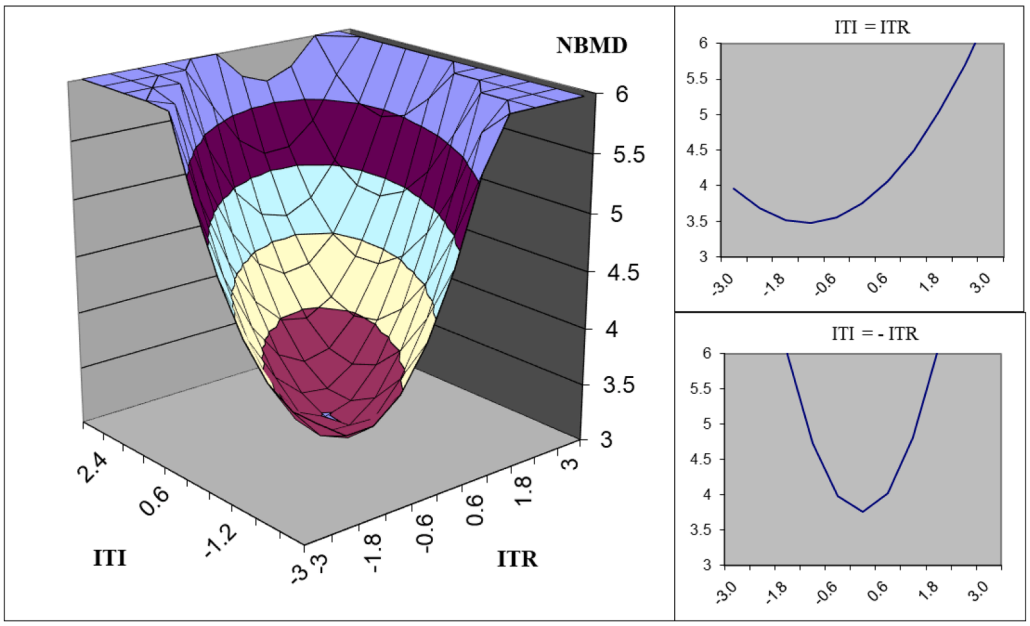


Fig. 1. Relationship between structural IT capabilities and NBMD.

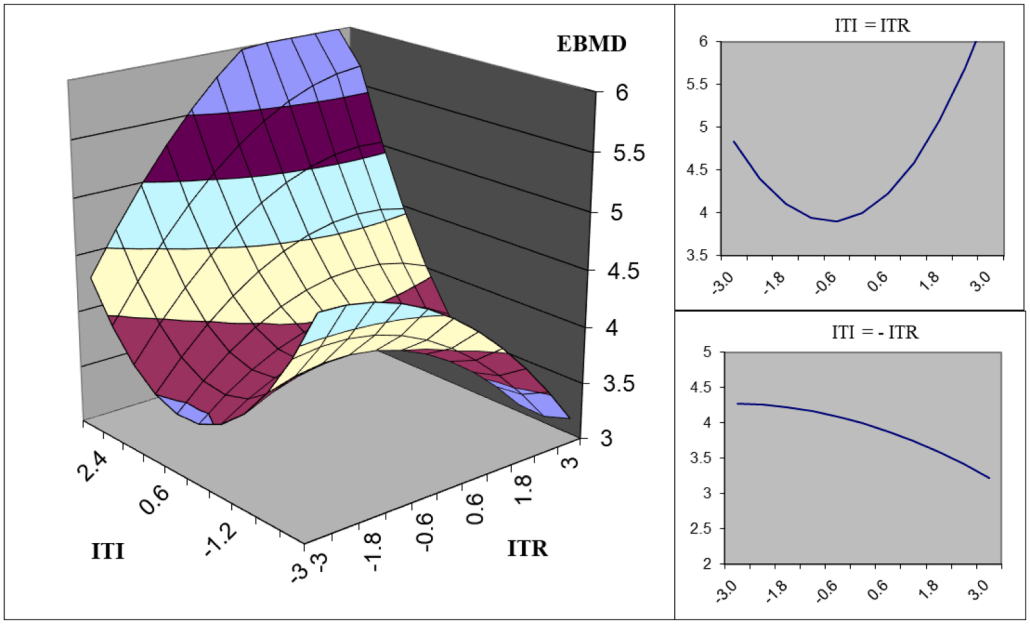


Fig. 2. Relationship between structural IT capabilities and EBMD.

linear relationship. Fig. 1 shows that when IT reconfiguration and integration efforts are balanced, the impact on NBMD always decreases and then increases as the total balanced effort increases, showing a U-shaped relationship. These results suggest that to maintain a high NBMD, a balanced effort of IT reconfiguration and integration needs to be low or high rather than moderate. Upon examining the situation when they are unbalanced, we find that the proportional balance effect of IT reconfiguration and integration has a significant curvature with a nonsignificant slope ($\beta = 0.710$, 90 % CI [0.156, 0.994]). This result suggests that a higher proportion of effort in either IT reconfiguration or integration will significantly boost NBMD. Thus, H3b considers only half of the scenarios.

H4a and H4b predict that a proportional balance effort of IT reconfiguration and integration positively impacts EBMD. Similarly to the results for NBMD, the slope ($\beta = 0.270$, 90 % CI [0.163, 0.361]) and curvature ($\beta = 0.182$, 90 % CI [0.128, 0.271]) of the response surface along the perfect balance line appear positive and significant. This result suggests that EBMD is better when balance effort is high. Therefore, H4a is validated. In addition, as with the results of the NBMD test, when IT reconfiguration and integration efforts are balanced, the effect on EBMD always decreases and then increases as the total balanced effort increases, again showing a U-shaped relationship. We then find that the proportionally balanced effect of IT reconfiguration and integration has a significant negative slope with a nonsignificant curvature ($\beta = -0.175$, 90 % CI [−0.312, −0.026]). This result suggests that enterprises should give IT integration a higher proportion of effort to better drive EBMD; thus, H4b is supported.

5. Discussion

The rapid advance of digital technology compels enterprises to design IT-based business models to gain a competitive advantage [71, 87]. The literature has emphasized the role of intra-organizational IT capabilities in BMD but has not yet given adequate attention to inter-organizational IT capabilities. To fill this research gap, this study uses the KBV to show that inter-organizational IT capabilities, such as structural IT capabilities, have the potential to enhance enterprises' capabilities to find and absorb external knowledge, thereby supporting BMD. Based on data from 180 manufacturing enterprises in China, our analysis yields the results in Table 4. These findings indicate that IT reconfiguration significantly promotes the beneficial effect of NBMD, and IT integration significantly promotes the beneficial effect of EBMD. These findings extend the literature on enterprise IT capabilities supporting BMD, especially inter-organizational IT capabilities.

This study has delved deeper into how a proportional balance of efforts toward IT reconfiguration and integration impacts NBMD and EBMD. The results in Table 6 show that when effort is evenly balanced between IT reconfiguration and integration, higher total effort supports NBMD and EBMD more effectively than low-level effort. Furthermore, as the evenly balanced effort of IT reconfiguration and integration grows, both NBMD and EBMD show a trend of first decreasing and then increasing, suggesting that blindly increasing IT reconfiguration and integration may be detrimental to BMD. We also provide the results for scenarios in which the IT reconfiguration and integration efforts are unbalanced. The results of response surface analysis indicate that any alteration in the high proportion of IT reconfiguration and integration efforts can promote NBMD. However, for EBMD, it is crucial to concentrate on increasing the proportion of IT integration to achieve maximum improvement. The above results reveal that striving for a perfect balance in IT reconfiguration and integration can be advantageous, but it is also necessary to explore the optimal proportional balance between them to achieve the optimal levels of NBMD and EBMD.

The direct effect hypotheses regarding IT reconfiguration and integration did not receive complete support. This discovery can be attributed to the difference in resource requirements between NBMD and EBMD. For example, research by Hu and Chen [41] highlights that NBMD requires creating connections and collaborating with new partners, whereas EBMD requires enterprises to be able to leverage their existing partnership networks fully. This corresponds with the concepts of IT reconfiguration and IT integration. The former significantly enhances the potential of enterprises to expand their new partner base and establish connections with new partners by building highly compatible enterprise information systems [97]. The latter facilitates the exchange of system information with existing partners, enabling the timely discovery of new business opportunities and cooperation possibilities. These arguments may elucidate why this study did not find support for a positive impact of IT reconfiguration on EBMD or a positive impact of IT integration on NMBD.

Table 6
Tests of response surface.

	NBMD Coefficient	90 % CI	EBMD Coefficient	90 % CI
Perfect balance line (ITI = ITR)				
Slope	**0.421**	[0.294, 0.544]	**0.270**	[0.163, 0.361]
Curvature	**0.163**	[0.022, 0.252]	**0.182**	[0.128, 0.271]
Perfect imbalance line (ITI = − ITR)				
Slope	0.033	[−0.217, 0.269]	**−0.175**	[−0.312, −0.026]
Curvature	**0.710**	[0.156, 0.994]	−0.027	[−0.303, 0.177]

In addition, we predict that when the effort put toward IT reconfiguration is greater than that put toward IT integration, there will be a better promotion effect on NBMD. However, our results indicate that NBMD is promoted irrespective of whether IT reconfiguration or IT integration receives the greater effort in an unbalanced situation. In the polynomial regression results, we observe a significant interaction between IT reconfiguration and IT integration for NBMD. This indicates that for enterprises to maximize NBMD, these two types of IT capabilities must coexist. The situation for EBMD is different, and polynomial regression does not reveal interaction between IT reconfiguration and IT integration. Therefore, for EBMD, it is not mandatory to balance IT reconfiguration and integration. EBMD can be improved mainly through the effects of IT integration.

It should be noted that NBMD and EBMD do need IT reconfiguration and integration to maintain a competitive advantage, which is not contradictory. For example, knowledge discovery and transaction innovation achieved through IT reconfiguration still need IT integration to complete knowledge storage, transmission, and transformation. The key issue is that to achieve the best results, the ideal effort proportions toward IT reconfiguration and IT integration may not necessarily be perfect balance.

5.1. Theoretical contributions

In this study, we unveil three major contributions. First, it is one of the few studies to examine inter-organizational IT capabilities as business model antecedents. While past studies have scrutinized the role of IT capabilities as business model enablers, almost all of these studies have considered intra-organizational IT capabilities, such as IT management capabilities and IT human capital [15,36,98]. Since the business model is a framework for business transactional activities across organizational boundaries [1,71,79], this presupposes that we need to investigate the inter-organizational IT capability perspective. Our study not only provides empirical evidence on the impact of inter-organizational IT capabilities, that is, structural IT capabilities, on business models but also highlights such IT capabilities as knowledge enablers that ultimately promote business model design.

Second, we contribute by deepening the understanding of the relationship between IT capabilities and BMD by introducing a proportional balance perspective rather than considering only equal balance. Prior research has recognized the importance and provided insights into the balance (and imbalance) of IT capabilities to drive BMD [15], but to our knowledge, this is the first study to explore how the direction of this proportion can be configured to optimize a business model. We propose that strategically allocating resources for IT reconfiguration and integration is critical for improving NBMD and EBMD effectively. The findings suggest that enterprises have multiple options—either a high proportion of IT reconfiguration or a high proportion of IT integration—to improve the novelty of BMD. This way of allocating resources does not apply to EBMD; enterprises need to align their resources toward IT integration to maximize EBMD effectiveness. Our study uncovers findings in this area that deepen the scholarly understanding of the relationship between IT and BMD.

Third, this study enriches the literature on KBV by investigating the impact of IT reconfiguration and integration capabilities on enterprise knowledge management from a proportional balance perspective. Existing literature has demonstrated the direct effect of IT on knowledge management extensively [9,21,37,47,50,81,91]. However, given the limited resources of enterprises, it is not clear how the proportional balance of different IT capabilities will affect knowledge acquisition and assimilation, and the related research is not yet sufficient. Since the external environment is an important source of knowledge [2], this study provides an in-depth discussion of how the proportional balance of IT reconfiguration and integration will affect the process of knowledge management. We find that firms can develop IT reconfiguration and IT integration in the form of proportional balance to drive

inter-organizational knowledge acquisition and assimilation, which ultimately contributes to the prosperity of BMD.

5.2. Managerial implications

Our research also has important implications for business managers. First, enterprises should attach importance to information system construction, especially structural IT capabilities. As the business environment develops rapidly, the relationship between enterprises becomes closer and closer, and it becomes increasingly difficult to achieve a solid competitive advantage alone [70,71]. The rapid development of information technology provides an opportunity to solve these problems. Enterprises can rely on inter-organizational information system to cooperate digitally with business partners. Our research points out the importance of inter-organizational IT capabilities to BMD, especially structural IT capabilities, which help enterprises acquire and integrate external knowledge, assimilate it into corporate knowledge, and promote the development of business models.

Second, companies should be extremely cautious in adopting balanced IT capabilities to promote BMD. According to our findings, both low and high overall balance efforts can produce positive results. Business managers should note that although low balance efforts are beneficial, they may not ensure the sustainability of a company's dominant position [15]. Therefore, in the early stages of business model development, corporate executives should make sustainable IT investments and balance IT reconfiguration and integration. Doing so can not only maintain a relatively high level of NBMD and EBMD in the early stages, but also lay the foundation for a subsequent transition. During the transition period, corporate managers need to pay attention to the company's resource reserves and adopt a project bimodal or enterprise bimodal approach [58] to allocate resources for IT reconfiguration or IT integration.

Third, business managers must take a proportional balance approach to investing in IT reconfiguration and integration in accordance with the company's goals. While the potential benefits of keeping IT reconfiguration and integration in balance are enormous, managers must also give due consideration to the rational development of internal resources. Underutilization of resources or overinvestment in a particular IT capability may not be conducive to an organization's knowledge management. Therefore, we do not recommend that companies simply and brutally separate the inputs of two IT capabilities. Companies need to give corresponding efforts to IT reconfiguration and integration at the same time. The proportion of effort allocated depends on the company's goal, that is, NBMD or EBMD. Our research results show that to achieve a high level of BMD, pursuing a high level of balance between IT reconfiguration and integration may not be the best choice, and that focusing on either IT reconfiguration or integration strategies independently is also beneficial to BMD.

5.3. Limitations and future research

Although our study contributes to the BMD and information systems literature, limitations remain. First, the sample of this study was selected from manufacturing enterprises in the Yangtze River delta region of China, and this sample may not facilitate the generalization of the findings. Specifically, despite controlling for firm size in our data analysis, we may not be able to eliminate the sample selection bias. Therefore, we recommend that future studies generalize the findings by comparing data from samples in different countries with varying firm sizes, which may reveal new insights. Second, because of the 2015 release of the Made in China 2025 policy, our sample data were collected from 2016 to 2017, but there is no doubt that collecting more recent data can greatly enhance the timeliness of our study. Third, this study uses a cross-sectional study design, which may lack information to determine causality, making it difficult to capture information on the evolution of business models. Therefore, we suggest that future studies deepen the understanding of this issue through longitudinal studies. Finally, future research could further explore the boundary conditions of the balanced configuration to deepen the understanding of this issue.

CRediT authorship contribution statement

Feiyu Wang: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Jibao Gu:** Supervision, Project administration, Funding acquisition. **Along Liu:** Writing – review & editing, Methodology, Formal analysis, Conceptualization.

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